

Assignment 2

1. Consider the full time-dependent Green's function in slab geometry, with

$$Q(x, t) = \delta(x)\delta(t).$$

For this problem, there exists a general relation between the solution in a purely scattering medium, $c = 1$, and the solution for a general value of c :

$$\psi(x, \mu, t; c) = ce^{-(1-c)t}\psi(cx, \mu, ct; 1).$$

Prove this relation by substituting the Ansatz into the time-dependent Boltzmann equation.

2. In the paper

J. C. J. Paasschens, "Solution of the time-dependent Boltzmann equation," *Physical Review E*, Vol. 56, No. 1, July 1997.

the following highly accurate solution for $c = 1$ is given for a spherical source:

$$\phi(r, t; 1) \simeq \frac{e^{-v\Sigma_t t}}{4\pi r^2} \delta(r - vt) + \frac{(1 - r^2/v^2 t^2)^{1/8}}{(4\pi vt/(3\Sigma_t))^{3/2}} e^{-v\Sigma_t t} G\left(v\Sigma_t t \left[1 - \frac{r^2}{v^2 t^2}\right]^{3/4}\right) \Theta(vt - r),$$

where

$$G(x) = 8(3x)^{-3/2} \sum_{N=1}^{\infty} \frac{\Gamma(\frac{3}{4}N + \frac{3}{2})}{\Gamma(\frac{3}{4}N)} \frac{x^N}{N!} \simeq e^x \sqrt{1 + \frac{2.026}{x}}.$$

Using the relation shown in class between the planar and spherical Green's functions, derive the corresponding planar solution.

Then, using the result of Question 1, obtain the exact solution for

$$c = 0.6, 0.8, 1, 1.2, 1.5.$$

Present the solution for

$$t = 1, 2, 3, 4, 7, 15.$$

Assume dimensionless space and time throughout this problem, i.e.,

$$\Sigma_t = 1, \quad v = 1.$$

3. (a) Add the classical diffusion solution to the plots obtained in Question 2.
 (b) Add the asymptotic diffusion solution to the plots obtained in Question 2.
 (c) Write a time-dependent numerical code for the diffusion equation, using both the classical diffusion approximation and the asymptotic diffusion approximation.
 Plot the numerical solutions together with the exact solutions obtained in Question 2.
 (d) For each value of c and t , compute the relative error of the classical and asymptotic diffusion approximations with respect to the exact solution and discuss the observed trends.